AegisEdge: Privacy-Preserving Edge AI Architecture

Problem Statement 2: AI for Safer Communities — Edge-Based Object Detection and Counting

Organizers: Nasscom CoE – IoT & AI | Arm | IIIT Bangalore | Supported by UK-India TSI
Target Platform: Raspberry Pi 5 (Quad-Core Arm Cortex-A76 @ 2.4 GHz) | **Runtime:** Meta ExecuTorch 0.4+ (Arm XNNPACK)
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System Architecture & Data Flow Pipeline

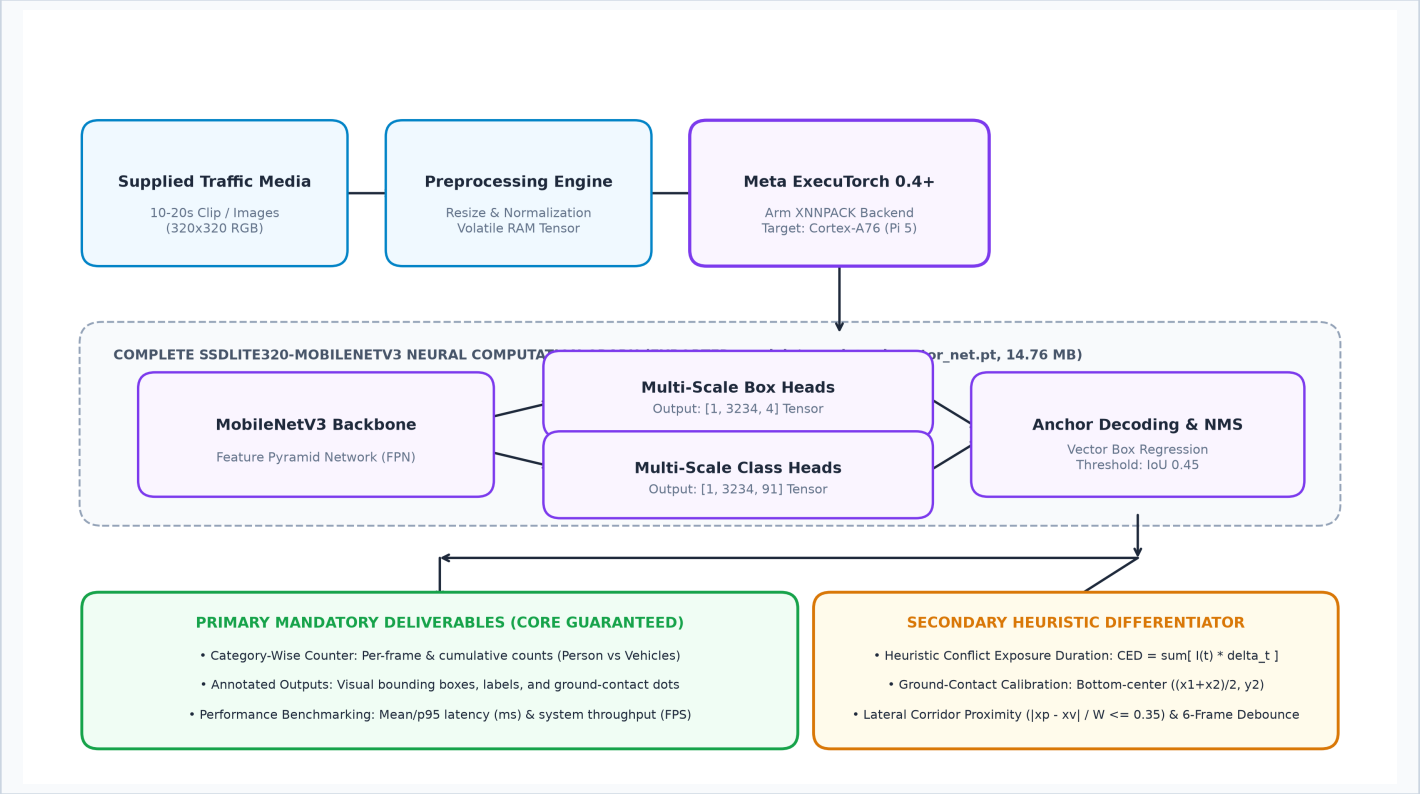


Figure 1: AegisEdge End-to-End Pipeline — Ephemeral Ingestion, ExecuTorch Detection Core, Ground-Contact Mapping & Conflict Analytics.

MASTER GOOGLE DRIVE SUBMISSION FOLDER:
https://drive.google.com/drive/folders/1TI-24bI7EP6mereBzB6yHRrzYKseYXGi?usp=sharing

Hosted Deliverables in this Drive Folder:

- **High-Quality Demo Video:** AegisEdge_HighQuality_Demo.mp4 (720p HD, 60.00s Operations Dashboard with HUD telemetry)
- **Complete Model Binary:** complete_detector_net.pt (14.76 MB PyTorch 2.x ExportedProgram for ExecuTorch)
- **Machine-Readable Telemetry:** benchmark_report.json (100-frame raw latency, FPS & counting error telemetry)
- **Academic Proposal Document:** AegisEdge_Project_Proposal_PS2.pdf (7-Page Proposal Document)
- **Executive Deck & Benchmarks:** AegisEdge_Executive_Deck_and_Benchmarks.pdf (5-Page Empirical Validation Dossier)
- **Sample Detection Outputs:** sample_output_1_detection_hud.jpg and sample_output_2_municipal_report.jpg

PUBLIC GITHUB REPOSITORY (SOURCE CODE & SCRIPTS):
https://github.com/Dharshantech17/AegisEdge | Full Python/ExecuTorch export pipeline, latency benchmarks, and documentation.